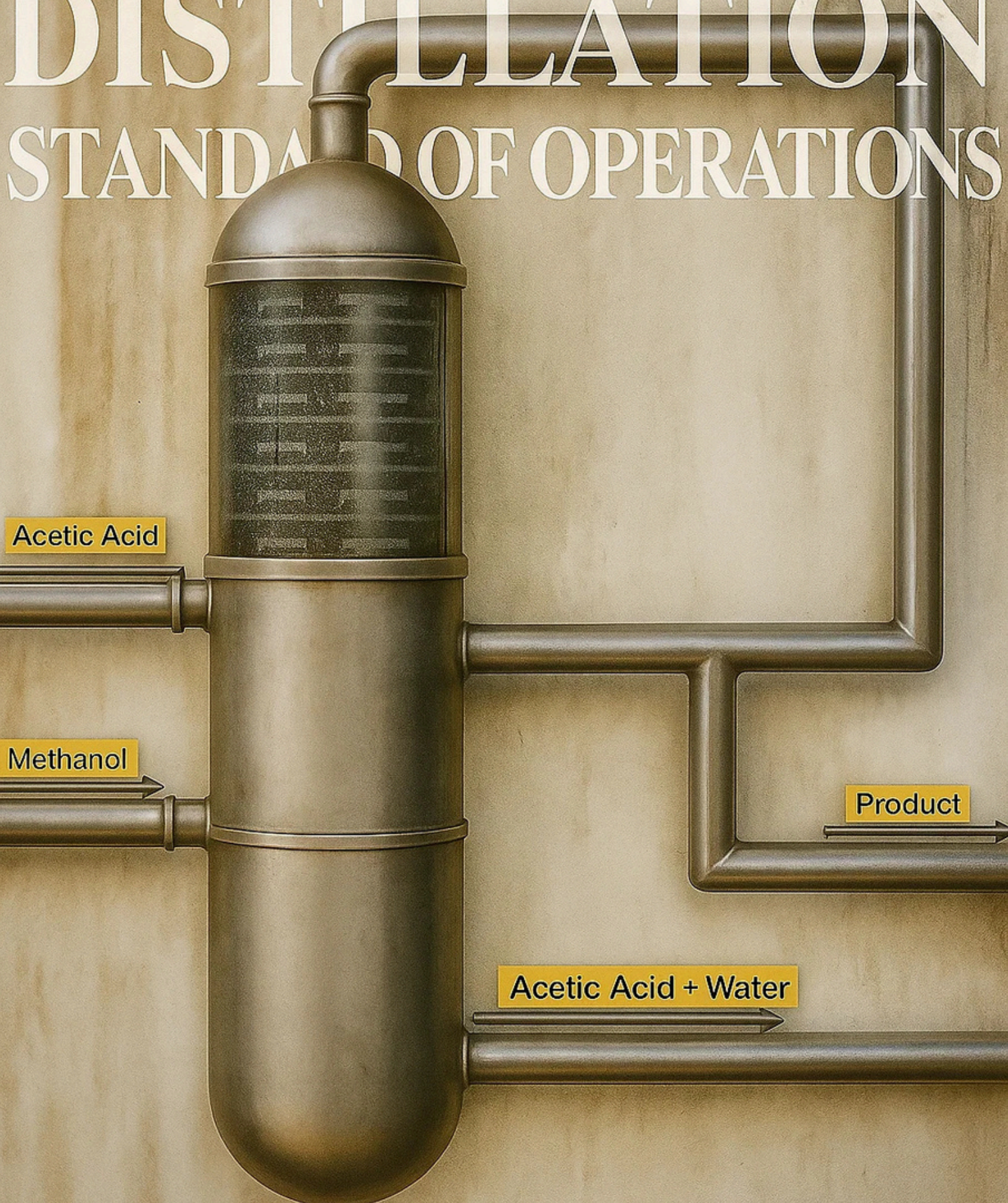


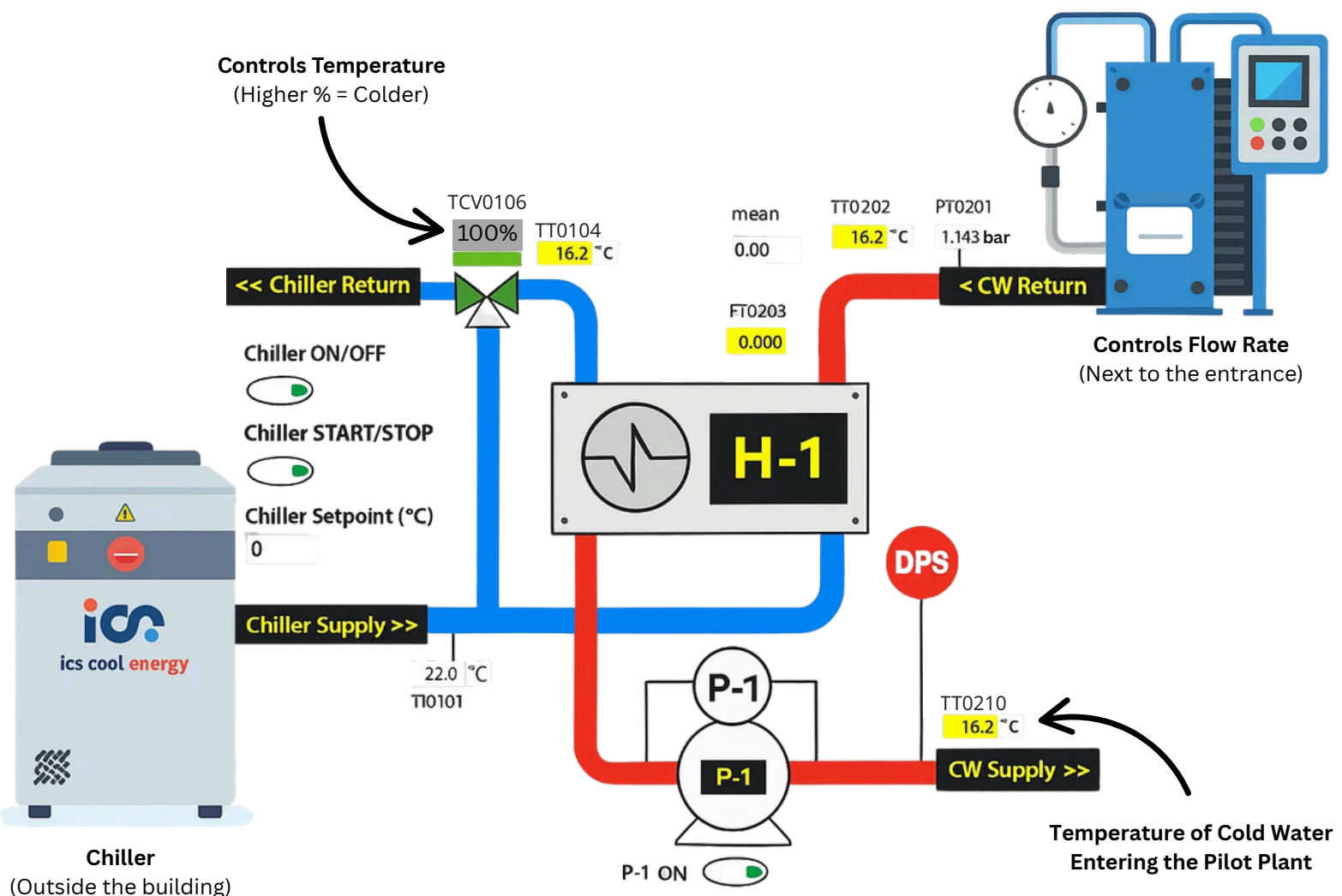
REACTIVE DISTILLATION STANDARD OF OPERATIONS



Cooling System (External)

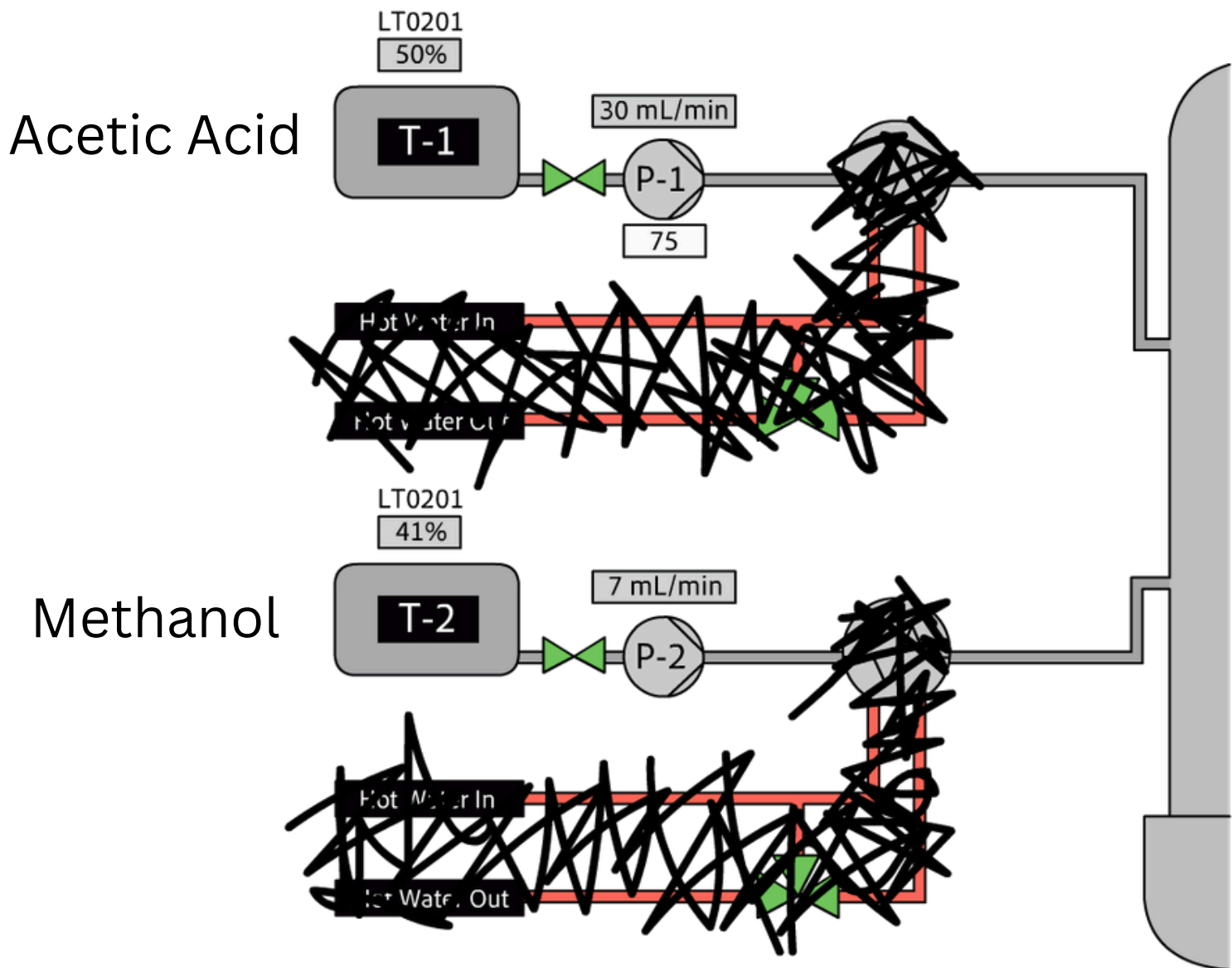
For safety reasons, it is impossible to turn on the heating system of the pilot plant until the cooling / condensation system is up and running.

There are two parts to the cooling system, one within the pilot plant, one external to it. The internal cooling system is dependent on the external one:



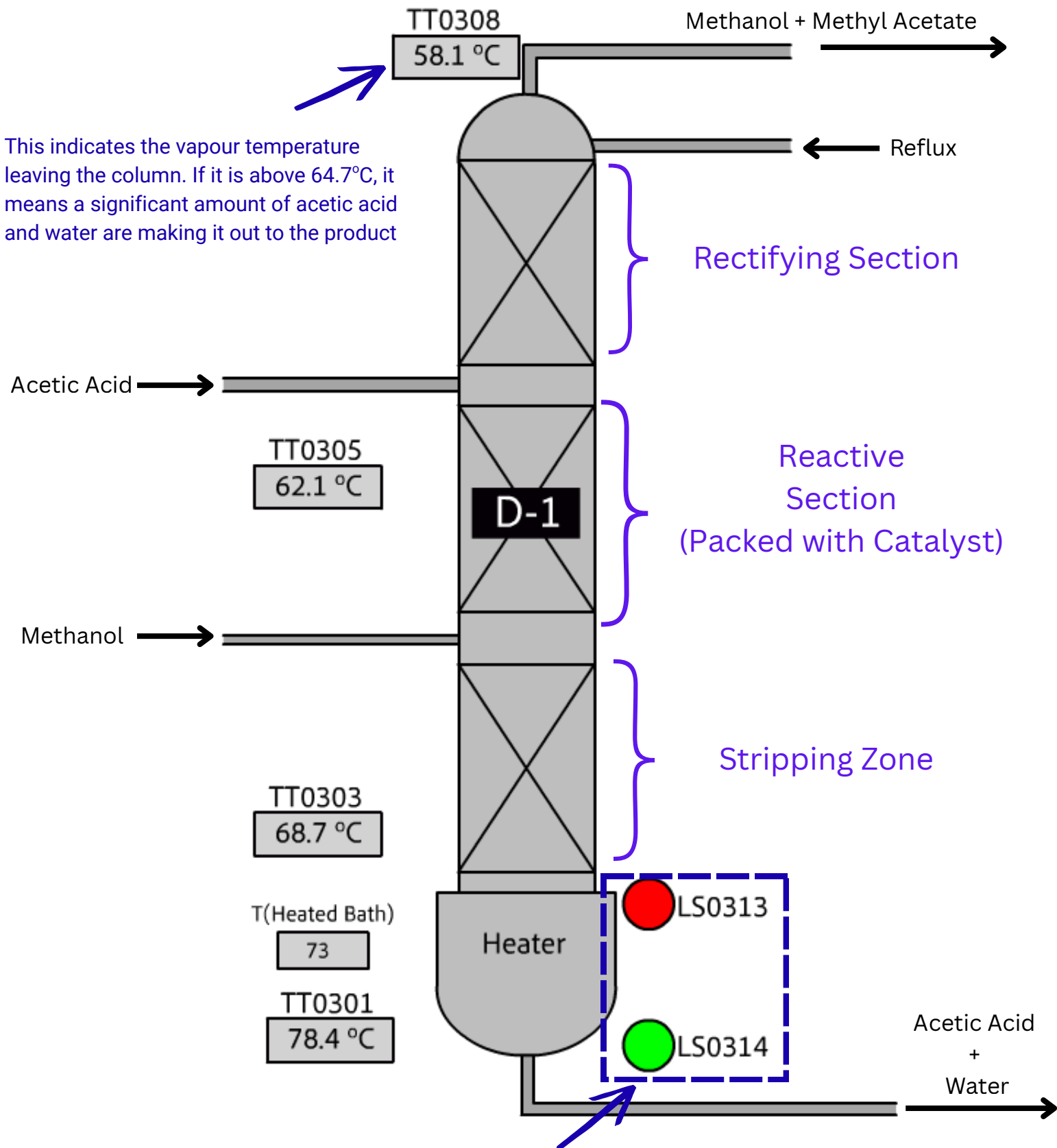
Feeding in Reactants

The pre-heat system for the reactants is not in operations, so ignore them. The only thing you can control now is the flow rate of the reactants



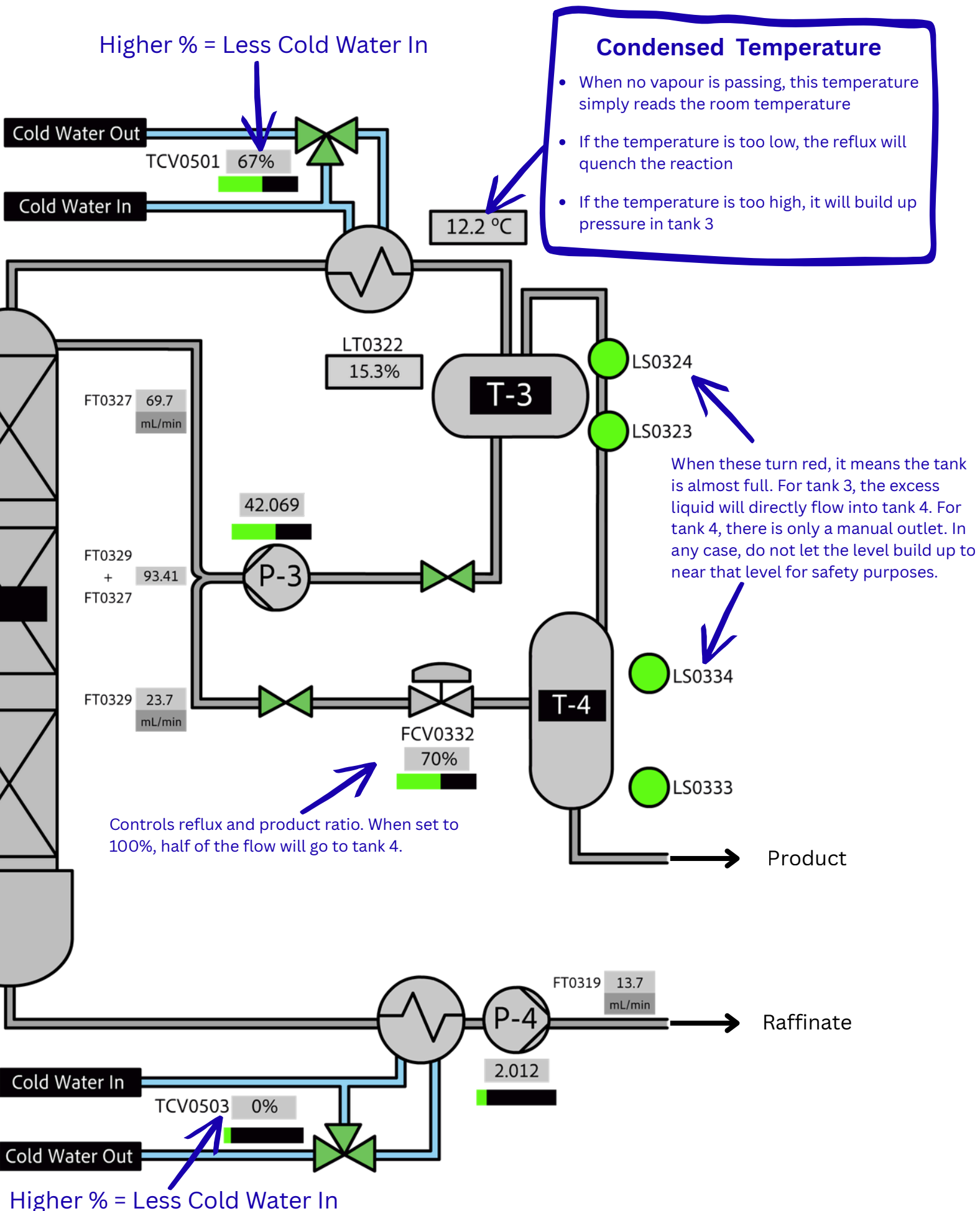
- The baseline flow meter reading is ≈ 2.1 mL/min, meaning it won't be zero even if nothing is flowing through it.
- Sometimes even when the pump is not on, the flowmeter may read a high flowrate. This is likely due to back flow of vapour from the column. Simply turn on the pump and drive the vapour away with liquid to return an accurate reading

Reaction and Distillation

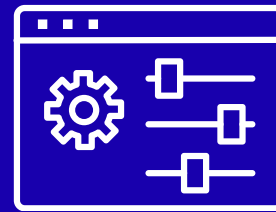


- When LS0313 is red, it means the liquid level reaches there. If it is green, the liquid level is below it
- When LS0314 is green, it means the liquid level reaches there. If it is red, there is no liquid left in the tank and the heating system will shut down immediately
- The volume between LS0313 and LS0314 is a pathetic 200ml of buffer volume, so the moment LS0313 starts flashing green and red, act quickly!

Product and Raffinate



Control



Front Panel Controllers Txy Density

9. FT0327+FT0329		10. Ratio		11. Composition		12. TT0301 - (Bang-bang Control)	
1. FIC0104	2. FIC0204	3. TT0105	4. TT0205	5. FT0319	6. TT0318	7. TT0320	8. LT0322

Flow Control:

- The flow you can control is FT0104 (acetic acid feed in), FT0204 (methanol feed in), FT0329+FT0327 (Total flow of product+reflux), FT0319 (Raffinate flow)

Column Inlet Flowrate - FIC0104 via P-1

Inactive ☒ PID ☐ Ratio (0104:0204) ☐

PV/CV 1
FIC0104

Setpoint 1
25.000

Actuator 1
P-1

Gain 1
0.50

MV 1
P-1 Power

Integral 1
0.100

Derivative 1
0.000

Heating Control:

- It is recommended to use time sequenced heating rather than PID

R-1 Bang-bang control via Sump Temperature (TT0301)

Inactive ☐ Active ☐ Timed Sequence ☒ BANG Pos. 0

Set Point °C	82.25	Time On	6
Upper Offset °C	0.050	Time Off	8
Lower Offset °C	0.050		

Cycle Status ☒

On Cycle Time -3.0531

Off Cycle Time 3.2